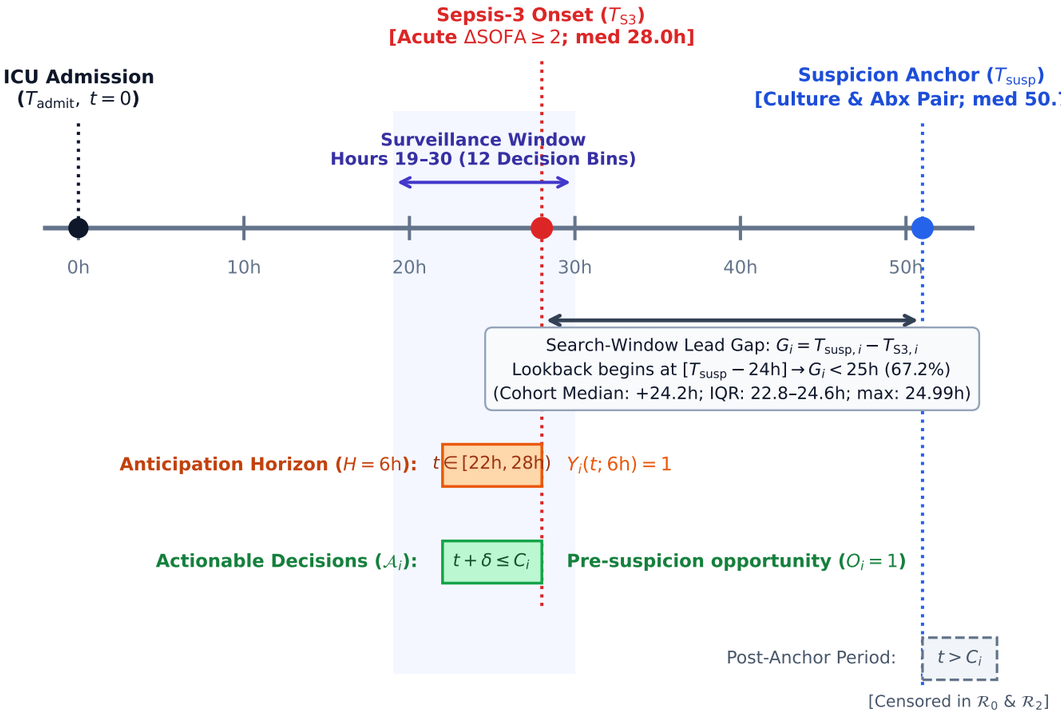
A. Operational Clocks & Actionable Decision Boundary



MATHEMATICAL RISK SET FORMULATION (§2.3)

- Evaluated Base Set ($\mathcal{R}$):
  $\mathcal{R} = \{(i,t) : i \in \mathcal{C},\ t \in \{19 \ldots 30\}\text{h},\ t < T_{S3,i}\}$ (97,700 decision windows across 8,254 stays; 4,070.8 pt-days)
- Anticipation Horizon ($H = 6h$):
  $Y_i(t; 6h) = \mathbf{1}\{t < T_{S3,i} \leq t + 6h\}$ (678 positive / 97,022 negative windows; base prevalence $\pi = 0.694\%$)
- Exploratory Hybrid Sets (Outcome-Stratified Restriction):
  $\mathcal{R}_\delta^{hybrid} = \{(i,t) \in \mathcal{R} : t + \delta \leq C_i\}$ with $C_i = T_{susp}$ for cases and $T_{Rx,start}$ for controls
  $\rightarrow \mathcal{R}_0^{hybrid}$: 70,192 windows (674 pos / 69,518 neg) | $\mathcal{R}_2^{hybrid}$: 70,129 windows (663 pos / 69,466 neg)
- Uniform Earliest-Prescription Sets (Prospective Operational Benchmark):
  $\mathcal{R}_\delta^{Rx} = \{(i,t) \in \mathcal{R} : t + \delta \leq T_{Rx,start,i}\}$ applied identically across all patients ($T_{Rx,start} \leq 3.83$h for cases)
  $\rightarrow \mathcal{R}_0^{Rx}$: 65,926 control windows (2,746.9 pt-days; $N_+ = 0$; discrimination mathematically non-estimable)
  $\rightarrow \mathcal{R}_2^{Rx}$: 65,877 control windows (2,744.9 pt-days; $N_+ = 0$; discrimination mathematically non-estimable)

B. Population Opportunity Attrition

